Problem 7

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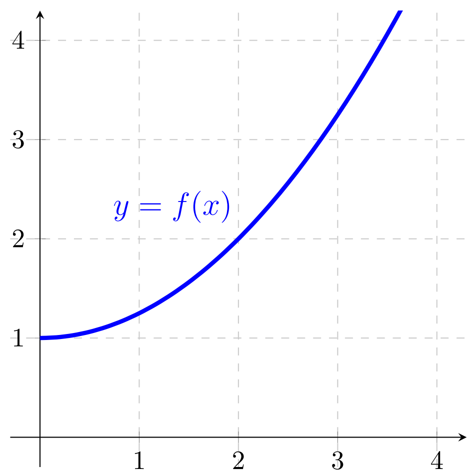
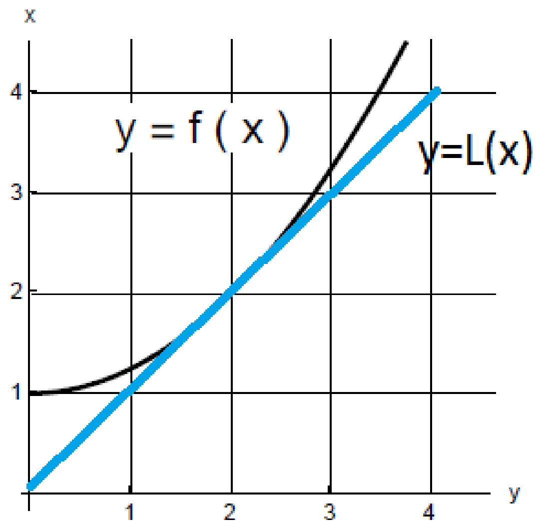
Problem 1 *The graph of a function f is given below.*



(a) *Given that $f'(2) = 1$, find the linear approximation L to the function f at $a = 2$. [2]*

Explanation. $L(x) = f(2) + f'(2)(x - 2) = 2 + (x - 2) = x$

(b) *Sketch the graph of L in the figure above.*

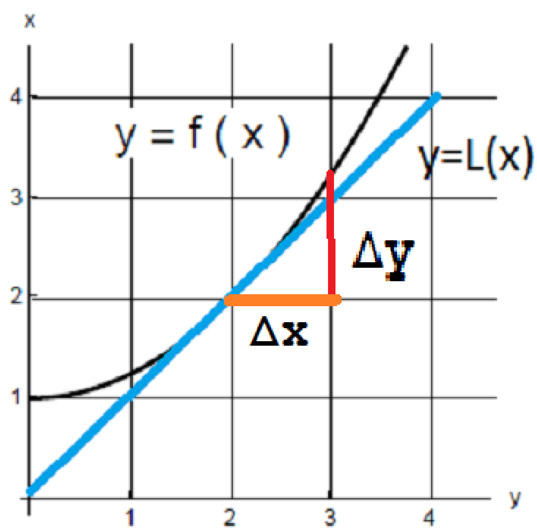


Explanation.

(c) *Use the linear approximation L to estimate the value of $f(3)$. Is this an underestimate or overestimate?* ***EXPLAIN****. [2]*

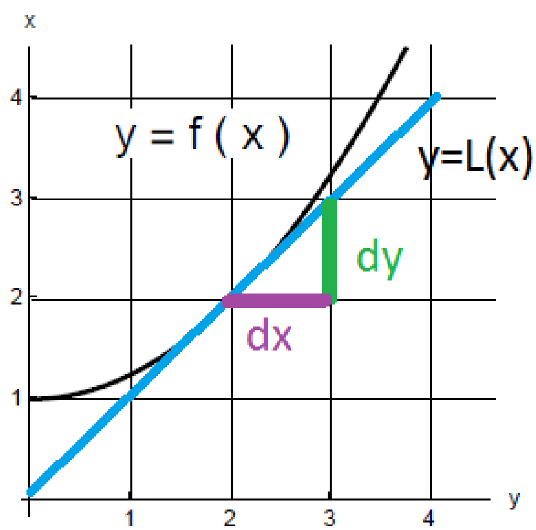
Explanation. $f(3) \approx L(3) = 3$. It is an underestimate because f is concave up on the interval $(2, 3)$.

(d) *When x changes from $a = 2$ to $a + \Delta x = 3$, the change in the* **function** *$y = f(x)$, Δy, is given by $\Delta y = f(a + \Delta x) - f(a)$. Draw and label Δy and Δx in the figure above.*



Explanation.

(e) *When x changes from $a = 2$ to $a + \Delta x = 3$, the change in the* **linear approximaton**, *dy, is given by $dy = L(a+\Delta x) - L(a) = f'(a)\Delta x$. Draw and label $L(x)$, dx and dy (differential) in the figure above.*



Explanation.